Biochemistry Cume

April 20, 2007

1. (25 points) When a tetrapeptide having an isoelectric point within the range 6.8 +/- 1.1 is digested with trypsin, two molar equivalents of a single digestion product are observed. Give plausible sequences for the tetrapeptide and digestion product.

2. (25 points) Draw the complete chemical structure for the most favorable ionization state of the tetrapeptide in problem 1 at pH 6.8, showing all atoms. (Do not be concerned about the stereochemistry.)

3. (25 points) Draw a suitable titration curve for the tetrapeptide in problem 1 over the pH range 0 to 14.

4. (5 points) How many different tetrapeptides can be made using only the amino acids glycine, L-alanine, L-valine and L-leucine?

5. (5 points) Repeat question 4 if each listed amino acid is used exactly once in a particular tetrapeptide.

6. (10 points) How many different tetrapeptides can be made using the 20 genetically encoded amino acids?

7. (5 points) Repeat question 6 if each genetically encoded amino acid is used no more than once in a particular tetrapeptide.